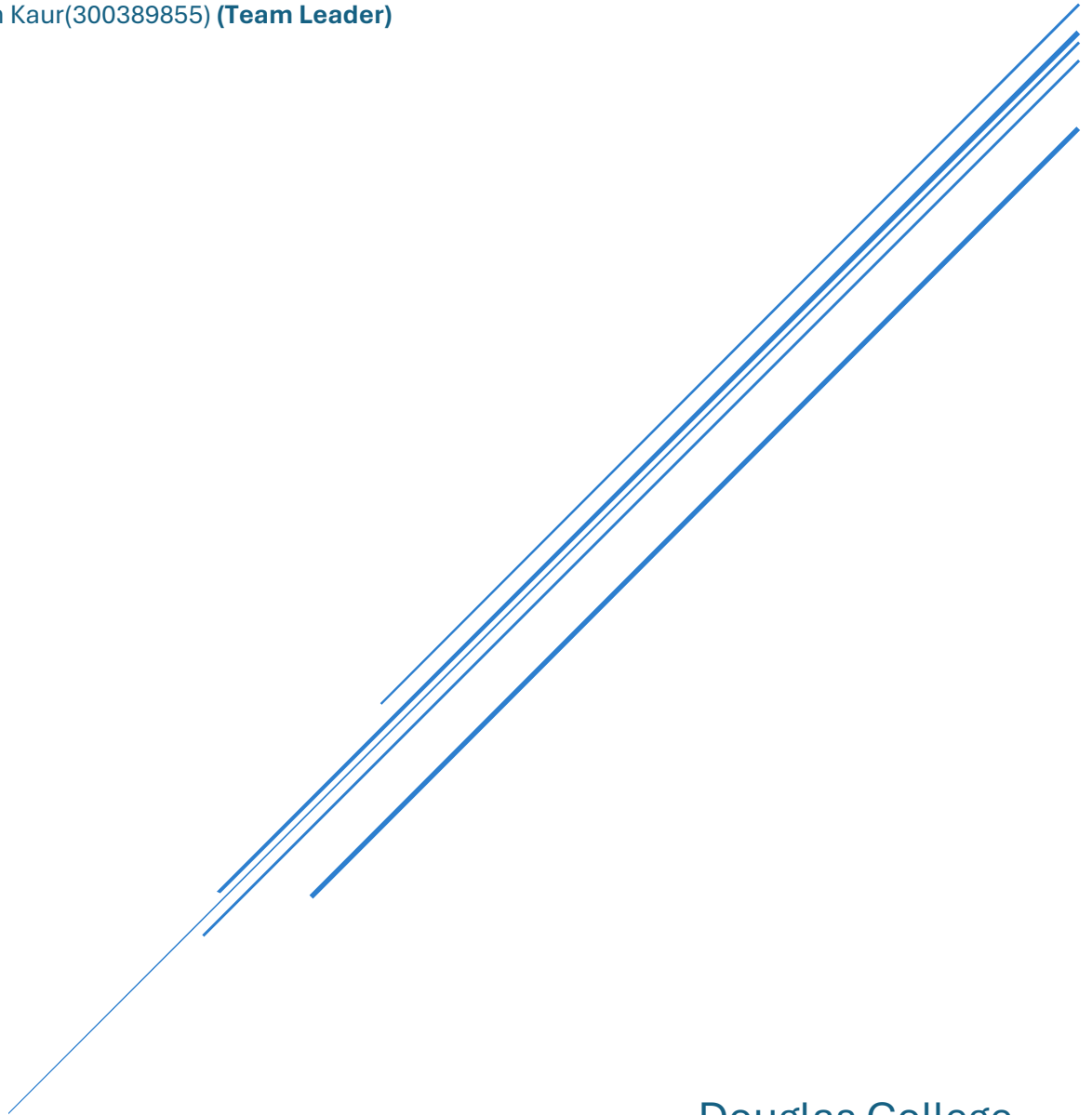


COMMUNITY EMERGENCY RESPONSE APP (CERA)

Applied Research Project

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Douglas College
CSIS 4495 – Fall 2025

COMMUNITY EMERGENCY RESPONSE APP (CERA) – PROJECT PROPOSAL

INTRODUCTION

Topic Overview:

Emergencies such as natural disasters, medical crises, or infrastructure failures often require rapid coordination between citizens, volunteers, and local authorities. Despite technological advances, many communities still lack an integrated platform for reporting incidents, dispatching volunteers, and tracking emergencies in real time. This project aims to develop a mobile-first solution, the Community Emergency Response App (CERA), to enhance the speed and effectiveness of community-based emergency response.

LITERATURE REVIEW:

Previous Work or Research:

Recent studies have explored mobile applications and systems for emergency management. A 2024 preprint on arXiv highlighted mobile-based platforms for disaster coordination emphasizing real-time updates, volunteer allocation, and resource optimization [1]. Another study in the *Journal of Engineering Science and Technology* (2021) examined community disaster apps focusing on communication and task management between citizens and authorities [2]. Moreover, a 2023 review in JETIR analyzed the effectiveness of crowd-sourced reporting and volunteer dispatch in emergency scenarios [3].

METHODOLOGY:

Overview of Approach:

This project will follow an applied software development methodology. It will involve iterative design and development of the mobile application using React Native, a RESTful backend with Express.js, and MongoDB for flexible data modeling. The system will include role-based access control for residents, volunteers, and coordinators.

Key modules include:

- Incident reporting with photo and geolocation attachments
- Volunteer dispatch with proximity-based notifications

- Live mapping of incidents and volunteers
- Offline caching to ensure functionality in low-connectivity zones (less priority)
- Admin dashboard for coordinators to approve volunteers and manage incidents

DATA COLLECTION:

Data will primarily consist of synthetic test incidents, user interaction logs, and simulated volunteer assignments to evaluate functionality. Additionally, wireframes and usability testing with volunteer peers or classmates will provide qualitative feedback on interface and workflow clarity.

DATA ANALYSIS:

- Quantitative analysis: Track response time between incident creation and volunteer assignment, number of incidents successfully dispatched, and offline sync success rate.
- Qualitative analysis: Usability evaluation through peer testing, interface intuitiveness, and workflow efficiency assessment.

Key Stakeholders

- Residents and community volunteers (end-users)
- Community coordinators or municipal staff
- NGOs and disaster relief organizations

LIMITATIONS:

- Testing relies on self created emergencies; real-world validation requires partnerships with municipalities.
- Backend hosted on basic or free version of cloud infrastructure may have performance limitations.

Expected Results and Implications

- A functioning mobile application that is capable of incident reporting, volunteer dispatch, and real-time mapping
- Usability-tested interface for residents and volunteers

- Documentation of implementation, lessons learned, and recommendations for future deployment
- Potentially if everything goes well it might be used in real world.

REFERENCES

1. Arxiv. (2024). Mobile Emergency Management Platforms for Community Response.
<https://arxiv.org/html/2407.08145v1#S2>
2. Jestec. (2021). Community Disaster Response Apps and Coordination.
https://jestec.taylors.edu.my/Speccial%20Issue%20ICITE2021/Special%20issue%20ICITE21_04.pdf
3. JETIR. (2023). Crowdsourced Emergency Reporting and Volunteer Dispatch.
<https://www.jetir.org/papers/JETIR2504B73.pdf>

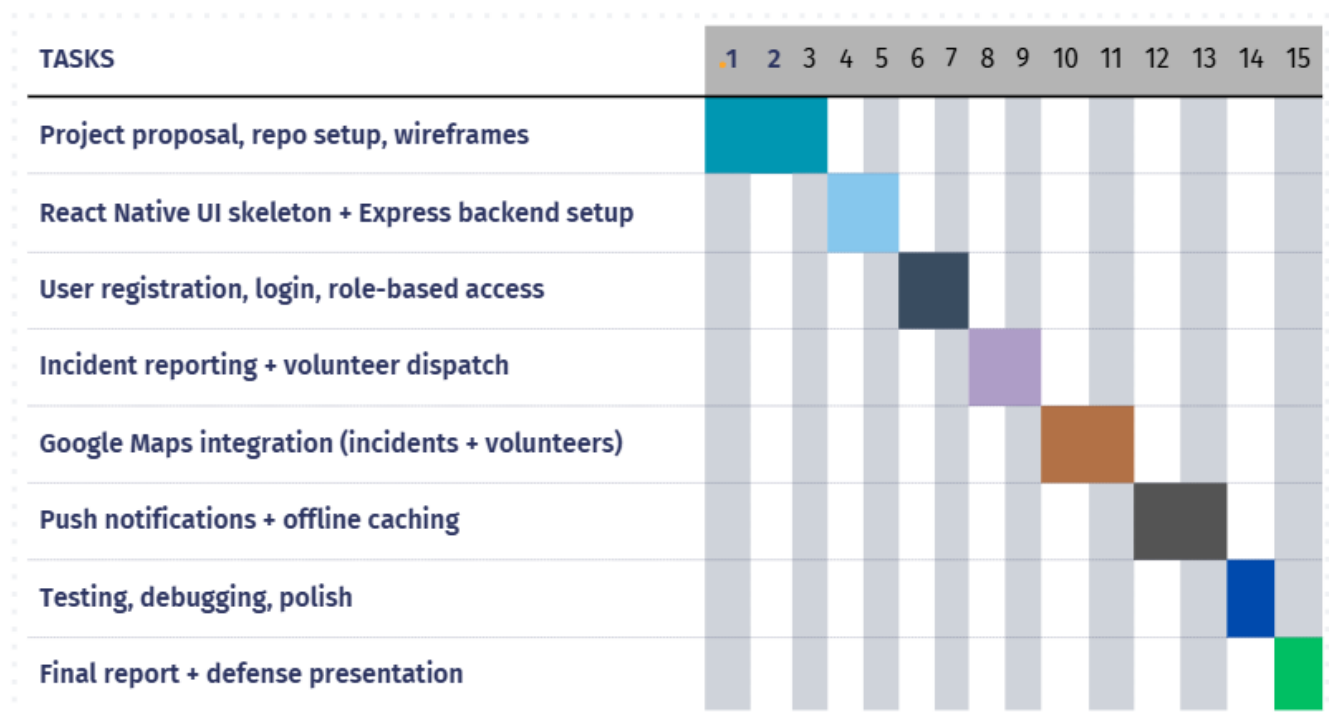
SCOPE OF WORK

1. **Frontend (React Native)** – UI for Android/iOS with user roles (residents, volunteers, coordinators).
2. **Backend (Express.js)** – RESTful API for user, incident, and dispatch management.
3. **Database (MongoDB)** – Stores users, incidents, and activity logs.
4. **Authentication (JWT)** – Role-based access control.
5. **Geolocation & Maps** – Google Maps API for incident and volunteer tracking.
6. **Notifications** – Push notifications for incident updates.
7. **Offline Support (Low Priority)** – Local caching for reporting and viewing incidents without internet.
8. **Admin Dashboard (Optional)** – Web-based management panel for coordinators.

GANTT CHART

Week	Task / Milestone	Deliverable
1–3	Project proposal, repo setup, wireframes	Proposal + GitHub setup
4–5	React Native UI skeleton + Express backend setup	Initial frontend/backend code
6–7	User registration, login, role-based access	Auth module
8–9	Incident reporting + volunteer dispatch	Reporting & dispatch module
10–11	Google Maps integration (incidents + volunteers)	Map view
12–13	Push notifications + offline caching	Notifications + offline support
14	Testing, debugging, polish	Midterm demo video
15	Final report + defense presentation	Final report + presentation

VISUAL PRESENTATION OF GANTT CHART



Detailed Implementation of Gantt Chart

Weeks 1–3: Project Preparation

User Stories

- As a project team, we need to define the scope, technical stack, and visualize the app so that we start with clarity and shared vision.

Tasks

- Create project proposal and requirements doc.
 - Set up GitHub repo with correct naming conventions/folders.
 - Build initial wireframes for mobile and admin interfaces.
 - Assign user roles (resident, volunteer, coordinator/admin).
-

Weeks 4–5: Basic App & Backend Skeleton

User Stories

- As a user, I want to see a welcome and login/register screen so I can access the app securely.
- As a coordinator, I want the backend structure in place to start storing user and incident data.

Tasks

- Scaffold React Native project, show basic navigation and authentication screens.
 - Setup Express.js backend with user and incident endpoints.
 - Define MongoDB schemas for users, roles, incidents.
 - Implement repository folder structure for code and docs.
-

Weeks 6–7: Registration, Login, Role-based Access

User Stories

- As a resident/volunteer, I want to register with my details and securely login, so I can use the app based on my role.

- As a coordinator, I want to approve or reject new volunteer registrations to ensure only qualified people join.

Tasks

- Develop registration and login screens (React Native).
 - Implement backend authentication with JWT and password hashing (bcrypt).
 - Role-based access/logins (restrict coordinator/admin views).
 - Admin approval flow for volunteers.
 - Security checkpoint: confirm password encryption, secure API routes, HTTPS setup (if deployed).
-

Weeks 8–9: Incident Reporting & Volunteer Dispatch

User Stories

- As a resident, I want to quickly report an emergency, attach location and optional photo, so responders know what's happening.
- As a volunteer, I want to receive notifications about nearby incidents and accept or decline assignments.
- As a coordinator, I want to verify incidents before dispatch to reduce false reports.

Tasks

- Build report-incident screen (form, photo, geotag).
 - Backend endpoint for new incident creation and volunteer dispatch logic.
 - Volunteer notification logic (task list/accept decline).
 - Incident filtering by type/status.
 - Coordinator approval step for incidents before dispatch.
-

Weeks 10–11: Maps & Geolocation Integration

User Stories

- As any user, I want to see incident locations and volunteer positions on a map, so I can visualize emergency activity in real time.
- As a coordinator, I want to assign volunteers based on region/proximity.
- As a coordinator, I want to see incident and volunteer stats, so I can monitor overall response levels.

Tasks

- Integrate Google Maps API in the mobile app.
 - Show incidents and volunteers as map markers.
 - Backend logic for location updates and proximity-based assignment.
 - UI for map filters (by type, urgency, etc.).
 - Basic analytics screen number of active incidents, volunteers available, completed responses.
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Weeks 12–13: Push Notifications & Offline Mode

User Stories

- As a volunteer, I need push alerts when new emergencies are reported, so I can respond immediately.
- As a user, I need the app to function (view, report) even with limited or no connectivity.

Tasks

- Implement push notifications for new incidents/tasks.
 - Local caching for offline support (incident list, reporting).
 - Sync logic (store offline, send when online).
 - UI for offline status alerts.
 - Security/privacy review checkpoint: confirm safe storage of location data and user details.
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Weeks 14: Testing, Bug Fixes, Midterm Demo/Video

User Stories

- As the project team, we need to verify all features, fix bugs, and demonstrate the core app functionality to stakeholders.

Tasks

- Write and run tests for major components (unit, integration, manual).
 - Bug-tracking and resolution across modules.
 - Prepare midterm demo and video showing core workflows.
 - Seed test/demo data for realistic emergency scenarios (fire, flood, medical).
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Weeks 15: Final Report, UI Polish, Final Defense Prep

User Stories

- As a user, I want a polished, accessible, and user-friendly app so I can participate confidently during an emergency.
- As a coordinator/instructor, I want to complete documentation of the project—including lessons learned and future recommendations.

Tasks

- Finalize UI/UX (clean, accessible, onboarding/tips).
- Accessibility checks (contrast, readable fonts, simple workflows).
- Write final project report and supporting documentation.
- Prepare and practice for defense presentation.
- Update GitHub and ensure all deliverables are present.

Additional Information / Future Updates

- The GitHub repository for this project will be setup after the project approval. Details like code structure, documentation, implementation progress will be updated later.
- A Kanban board will be set up for the project with detailed tasks, timelines which will be based on the Gantt chart.
- The main target is to implement the idea; if some feature or tasks are due to implement or dropped from the project, it will be communicated with professor right away.

PROJECT AGREEMENT:

Project Duration: – 15 Sept 2025 To 27-Nov-2025

Team Members:

1. Dilpreet Sandhu
2. Harleen Kaur

1. Agreement on Scope of Work

We, the undersigned team members, agree to work collaboratively on the development of the **Community Emergency Response App (CERA)**. The project scope includes:

Scope of Work:

- Design and implement a mobile application for residents, volunteers, and coordinators.
- Modules to include:
 - Incident reporting with photos and geolocation
 - Volunteer dispatch with proximity notifications
 - Live mapping of incidents and volunteers
 - Push notifications and offline caching (low priority)
 - Admin dashboard for coordinators
- Backend development with Express.js and MongoDB
- Authentication with JWT for secure, role-based access.
- Testing, debugging, and usability evaluation.
- Final report, documentation, and defense presentation.

Deliverables and Timeline:

Week	Task / Milestone	Deliverable
1–3	Project proposal, repo setup, wireframes	Proposal + GitHub setup
4–5	React Native UI skeleton + Express backend setup	Initial frontend/backend code
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2. Roles and Responsibilities

Team Members Agree To:

- Attend weekly meetings and participate in planning, discussion, and task completion.
- Complete assigned tasks on time and maintain code/documentation standards.
- Test individual components before integration.
- Update GitHub repository with all code and deliverables
- Communicate issues promptly to the team.

3. Meeting Frequency and Format

- **Frequency:** Weekly meetings every [Day] at [Time].
- **Format:** Online via [Platform] (Zoom, Google Meet) or in-person if necessary.
- **Agenda:**
 1. Review previous tasks and progress.

2. Discuss challenges and blockers.
3. Assign new tasks for the upcoming week.

4. Communication and Task Management

- Team will use WhatsApp for daily updates and quick questions.
- Any changes to scope, timeline, or deliverables must be agreed upon by all members and communicated to the professor.

5. Signatures

By signing this agreement, each team member confirms their understanding and acceptance of the project scope, responsibilities, and timelines.

Name	Signature	Date
Dilpreet Sandhu	Dilpreet Sandhu	15-Sept-2025
Harleen Kaur	Harleen Kaur	15-Sept-2025

This contract ensures **clarity, accountability, and smooth collaboration** for the successful completion of the CERA project.

WORK LOGS:

Dilpreet Sandhu:

Date	Task Completed	Hours Spent
4- Sept-2025	Created project group with Harleen. Explored different projects of Riipen and choose two of them 1.Glu 2.Weather Driver Decided to go with weather Driver after reading the description and discussed with group partner.	2 Hours
11-Sept-2025	Research more about the description of project to get the idea and explored many applications related to this topic.	1.5 Hours
13-Sept-2025	Had meeting on call with Harleen to know what exactly the requirements of projects. Juliet explained her about two projects but we both decide to go with none of them and instead of it we decided to create our own app from scratch. Explored various mobile and web applications for community services and volunteer coordination. Collected references and analyzed features to understand functionality and user interaction patterns.	2 Hours
14-Sept-2025	Had a group meeting on call to discuss various project ideas and, after discussion, decided to focus on developing a Community Emergency Response App.	2.5 Hours

	Started Documentation Wrote Literature Review, Limitations , Expected Results and Implications, Create Gantt chart	
15-Sept-2025	Wrote detailed implementation of Gantt chart with user stories	1.5 Hours

Harleen Kaur:

Date	Task Completed	Hours Spent
4- Sept-2025	Created Group with Dilpreet. Choose two projects of Riipen and decided to go with Weather Driver App as I wanted to create mobile app in react native.	2 Hours
7-Sept-2025	Explored the existing apps similar to weather driver app i.e. Drive Weather	1 Hour

13-Sept-2025	Had zoom meeting with Juliet and discussed about the projects. She offered two projects: 1. Customer Service Portal web app 2. Security Early Detection mobile app And then I researched on project. Discussed with Dilpreet about the projects and conclude to withdraw the request for ripen projects as the first project was out of scope for us as we want to create an app and first one was web app and second for large and complex for 2 members to complete within time limit. And decided to create our own app.	2.5 Hour
14-Sept-2025	Researched and brainstormed potential project ideas and discussed with Dilpreet and decided to go with his idea of CERA . Informed Juliet and Professor regarding the project withdraw. Started documentation Wrote Overview, methodology and data collection.	3 Hours
15-Sept-2025	Wrote Scope of work and create project agreement	1.5 Hours